

AS-AD Dynamics And Energy Price Shocks

Macroeconomics B
Lecture 6 (Chapter 19)

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Where We Are

- ▶ Lecture 4 gave us aggregate demand from IS, monetary policy, and the real interest rate.
- ▶ Lecture 5 gave us aggregate supply from price setting, costs, and expectations.
- ▶ Today combines both blocks into one dynamic AS-AD system.
- ▶ The question is practical: what happens when energy prices jump?
- ▶ We focus on intuition first, and use algebra only to keep the story disciplined.

Roadmap For Today

1. The 2022 energy shock as the motivating case
2. The AS-AD model in gaps
3. Persistence under static expectations
4. Demand shocks versus supply shocks
5. Policy choices after an energy supply shock
6. Current-event stress test: oil prices and the Strait of Hormuz

Learning Goals

After this lecture, you will be able to:

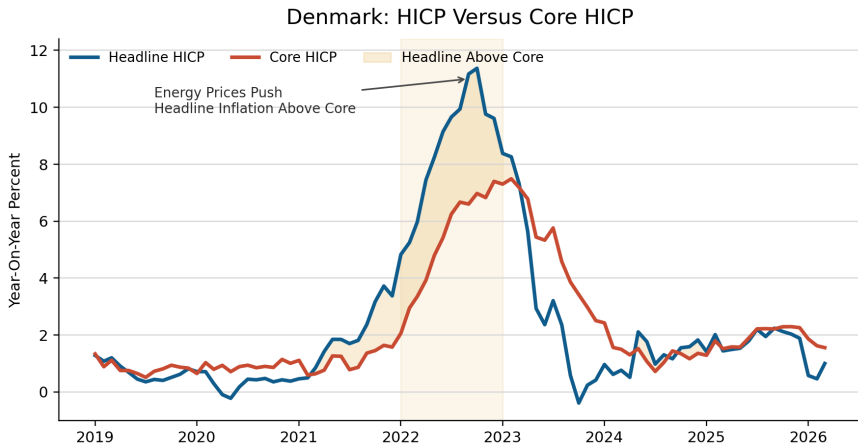
- ▶ derive the AD curve from IS, the real-rate definition, and the Taylor-type rule.
- ▶ explain how static expectations generate persistence after a shock.
- ▶ compute how $\beta = 1/(1 + a\gamma)$ governs convergence speed.
- ▶ distinguish demand shocks from supply shocks using output and inflation co-movement.
- ▶ map the 2022 euro-area energy shock into an adverse supply shock.
- ▶ assess how a stronger policy response changes the inflation-output trade-off.

Opening Case: Europe In 2022

- ▶ Energy markets tightened in 2021 and the shock intensified after Russia invaded Ukraine in February 2022.
- ▶ The IEA describes 2022 as a global energy crisis: natural gas prices reached record highs, electricity prices rose, and oil hit its highest level since 2008.
- ▶ Euro-area HICP inflation averaged 8.4 percent in 2022, up from 2.6 percent in 2021.
- ▶ The ECB states that energy prices were the most important component behind the rise in headline inflation.
- ▶ The policy question: should the central bank mainly fight inflation, cushion output, or balance the two?

Sources: IEA, "2022 Energy Crisis"; ECB Annual Report 2022.

Denmark: Headline And Core HICP



Sources: Eurostat HICP via FRED, series CP0000DKM086NEST and 00XFUNDKM086NEST. Core HICP excludes energy and unprocessed food.

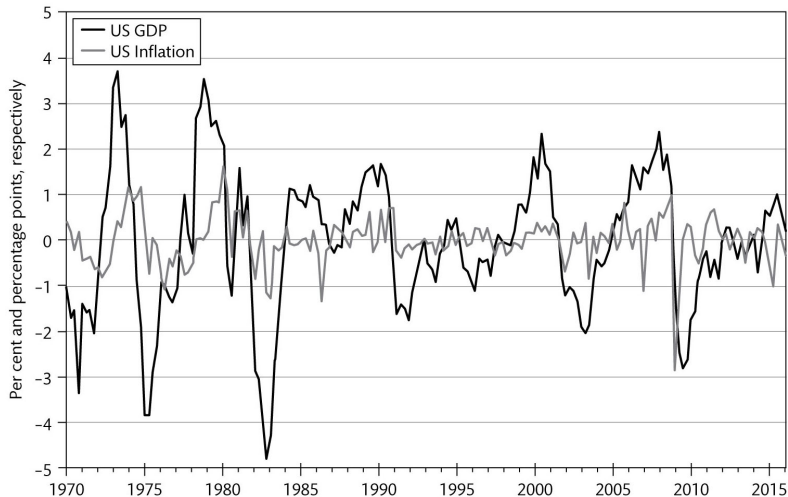
Why Energy Prices Are A Supply Shock

- ▶ Energy is an input into transport, heating, electricity, and production.
- ▶ Higher energy costs raise firms' marginal costs.
- ▶ Firms pass part of the cost increase into prices.
- ▶ Real purchasing power falls, so output demand may weaken too.

$$\pi_t = \pi_t^e + \gamma(y_t - \bar{y}) + s_t$$

- ▶ Energy cost shock: $s_t > 0$
- ▶ Inflation rises at a given output level.
- ▶ Output and inflation move in opposite directions on impact.

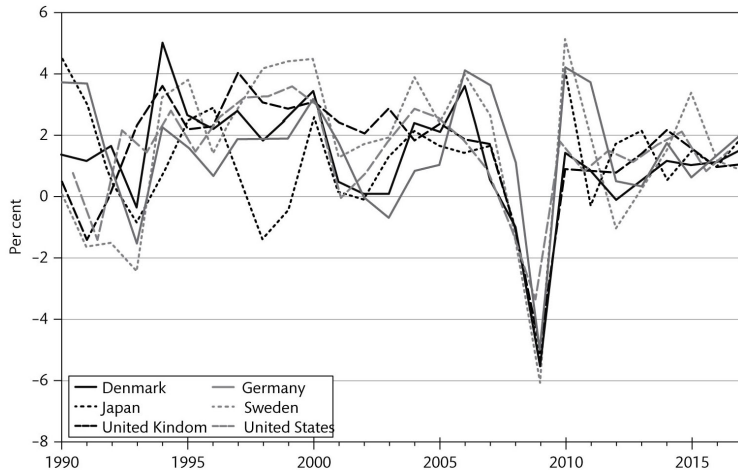
Business Cycles: Output Fluctuates Around Trend



Source: Sorensen & Whitta-Jacobsen, Chapter 19.

Shocks arrive suddenly, but output and inflation usually adjust over several periods.

International Co-Movement



Source: Sorensen & Whitta-Jacobsen, Chapter 19.

This example is based on the Global Financial Crisis. It illustrates international co-movement.

The Model Separates Impulse And Propagation

- ▶ An **impulse** is the initial disturbance: energy prices, fiscal shifts, confidence, risk premia, or productivity.
- ▶ **Propagation** is the model's internal adjustment after the impulse.
- ▶ Static expectations create inflation inertia: last period's inflation shapes current price setting.
- ▶ Monetary policy changes the slope of AD through the real-rate channel.
- ▶ The AS-AD model lets us ask: how large is the initial trade-off, and how fast does it fade?

Model Components

Demand Block

$$y_t - \bar{y} = \alpha_1(g_t - \bar{g}) - \alpha_2(r_t - \bar{r}) - \alpha_3(\pi_t - \bar{\pi}) + v_t$$

$$r_t = i_t - \pi_{t+1}^e$$

$$i_t^p = \bar{r} + \pi_{t+1}^e + h(\pi_t - \pi^*) + b(y_t - \bar{y})$$

$$i_t = i_t^p + \hat{\rho}_t$$

IS

Fisher equation

Taylor-type rule

Transmission to market rate

Supply Block

$$\pi_t = \pi_t^e + \gamma(y_t - \bar{y}) + s_t$$

AS

$$\pi_t^e = \pi_{t-1}$$

Static expectations

Demand Side: From Policy To AD

Start from the policy rule and the transmission equation:

$$\begin{aligned}i_t &= i_t^p + \hat{\rho}_t \\&= \bar{r} + \pi_{t+1}^e + h(\pi_t - \pi^*) + b(y_t - \bar{y}) + \hat{\rho}_t.\end{aligned}$$

Use the Fisher equation:

$$\begin{aligned}r_t - \bar{r} &= i_t - \pi_{t+1}^e - \bar{r} \\&= h(\pi_t - \pi^*) + b(y_t - \bar{y}) + \hat{\rho}_t.\end{aligned}$$

Expected inflation and $\bar{r}$ cancel. Insert this real-rate gap into IS:

$$y_t - \bar{y} = \alpha_1(g_t - \bar{g}) - \alpha_2[h(\pi_t - \pi^*) + b(y_t - \bar{y}) + \hat{\rho}_t] - \alpha_3(\tau_t - \bar{\tau}) + v_t.$$

Collect output terms:

$$y_t - \bar{y} = z_t - a(\pi_t - \pi^*)$$

where

$$a = \frac{\alpha_2 h}{1 + \alpha_2 b}, \quad z_t = \frac{\alpha_1(g_t - \bar{g}) - \alpha_3(\tau_t - \bar{\tau}) + v_t - \alpha_2 \hat{\rho}_t}{1 + \alpha_2 b}.$$

Derivation

z_t shifts AD; a governs how strongly inflation moves output through policy.

Taylor-Type Policy And The AD Slope

- ▶ If inflation rises above target, the rule raises the policy rate.
- ▶ A higher nominal rate raises the real rate when expected inflation does not move one-for-one.
- ▶ Higher real rates reduce demand.
- ▶ A larger inflation response h makes AD flatter in (y, π) space.

$$a = \frac{\alpha_2 h}{1 + \alpha_2 b}$$

$$y_t - \bar{y} = z_t - a(\pi_t - \pi^*)$$

Supply Side: Costs And Expectations

$$\pi_t = \pi_t^e + \gamma(y_t - \bar{y}) + s_t$$

- ▶ $\gamma > 0$: demand pressure raises marginal costs and inflation.
- ▶ s_t : cost-push disturbance, for example energy, markups, or productivity.
- ▶ Static expectations imply $\pi_t^e = \pi_{t-1}$.

$$\pi_t = \pi_{t-1} + \gamma(y_t - \bar{y}) + s_t$$

The AS-AD System In Gaps

Define

$$\hat{y}_t \equiv y_t - \bar{y}, \quad \hat{\pi}_t \equiv \pi_t - \pi^*.$$

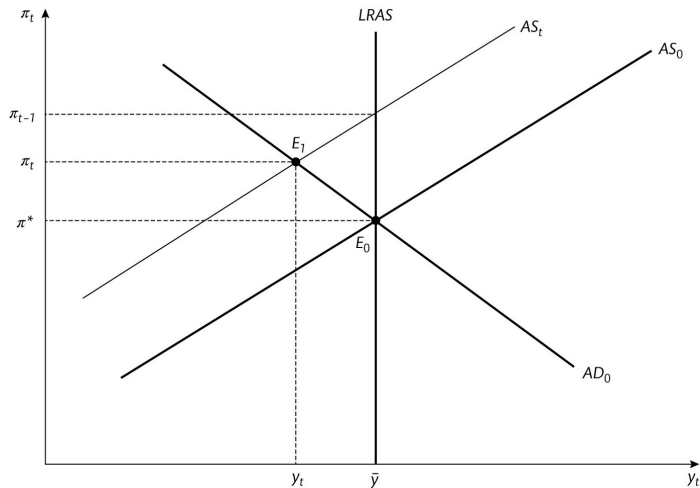
The compact AS-AD system is

$$\hat{y}_t = z_t - a\hat{\pi}_t$$

$$\hat{\pi}_t = \hat{\pi}_{t-1} + \gamma\hat{y}_t + s_t$$

- ▶ AD pins down demand at a given inflation gap.
- ▶ AS pins down inflation at a given output gap and inherited inflation.
- ▶ $a > 0$ is the AD sensitivity: larger a means inflation triggers a larger demand contraction through the real-rate channel.
- ▶ $\gamma > 0$ is the AS sensitivity: larger γ means output pressure translates more strongly into inflation.
- ▶ z_t is a demand impulse; s_t is a supply impulse.

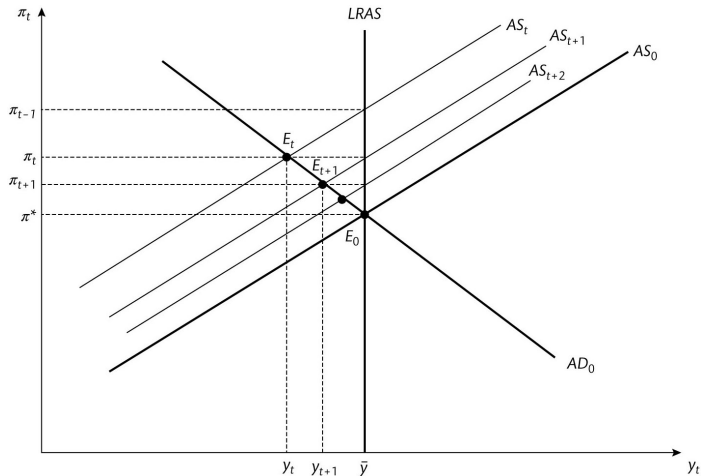
Graphical Equilibrium



Source: Sorensen & Whitta-Jacobsen, Figure 19.3.

Short-run equilibrium is the intersection of the current AD curve and the current AS curve.

Adjustment Toward Long-Run Equilibrium



Source: Sorensen & Whitta-Jacobsen, Figure 19.4.

With static expectations, AS shifts over time as inherited inflation changes.

Persistence: The Central Coefficient

Insert AD into AS:

$$\hat{\pi}_t = \hat{\pi}_{t-1} + \gamma(z_t - a\hat{\pi}_t) + s_t.$$

Collect inflation terms:

$$(1 + a\gamma)\hat{\pi}_t = \hat{\pi}_{t-1} + \gamma z_t + s_t.$$

Define

$$\beta \equiv \frac{1}{1 + a\gamma} \in (0, 1)$$

so that

$$\hat{\pi}_t = \beta\hat{\pi}_{t-1} + \gamma\beta z_t + \beta s_t$$

β tells us how much of last period's inflation gap carries into this period.

What Makes Adjustment Faster?

If there are no new shocks,

$$\hat{\pi}_t = \beta \hat{\pi}_{t-1}.$$

- ▶ A smaller β means faster convergence.
- ▶ a measures how strongly inflation moves demand through the real-rate channel.
- ▶ γ measures how strongly the output gap moves inflation through marginal costs.
- ▶ Since $\beta = 1/(1 + a\gamma)$, adjustment is faster when a or γ is larger.

Energy Shock Intuition In One Picture

- ▶ Period 1: energy costs rise, so AS shifts up.
- ▶ Inflation rises above target.
- ▶ Output falls below normal.
- ▶ Period 2 and after: inherited inflation keeps AS elevated.
- ▶ The economy returns gradually as inflation expectations adjust.

$$s_1 > 0, \quad s_t = 0 \text{ for } t \geq 2$$

$$\hat{\pi}_1 = \beta s_1 > 0$$

$$\hat{y}_1 = -a\beta s_1 < 0$$

$$\hat{\pi}_t = \beta^t s_1, \quad \hat{y}_t = -a\beta^t s_1$$

An adverse supply shock creates stagflation on impact.

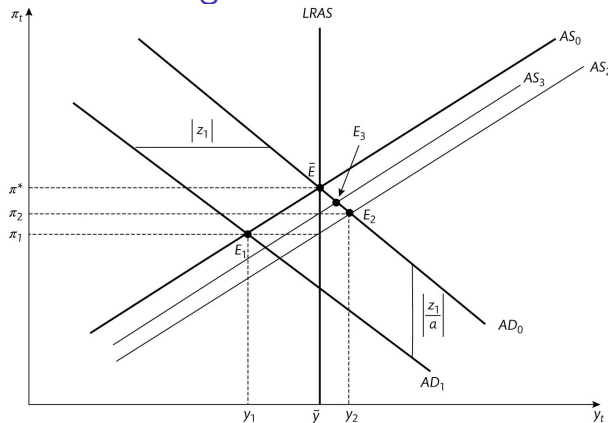
Temporary Demand Shock

Suppose $z_1 > 0$ and $z_t = 0$ for $t \geq 2$, while $s_t = 0$.

$$\hat{\pi}_1 = \gamma\beta z_1 > 0, \quad \hat{y}_1 = \beta z_1 > 0.$$

- ▶ Output and inflation rise together on impact.
- ▶ The model predicts positive co-movement.
- ▶ Fiscal expansions, confidence booms, or lower risk premia can look like this.
- ▶ Demand shocks are not the natural first description of the 2022 European energy episode.

Temporary Demand Shock: Figure



Source: Sorensen & Whitta-Jacobsen, Chapter 19.

A demand impulse shifts AD first and moves output and inflation in the same direction. AS moves after the first impact because static expectations imply $\pi_t^e = \pi_{t-1}$: higher inflation in period 1 is inherited by price setters in period 2.

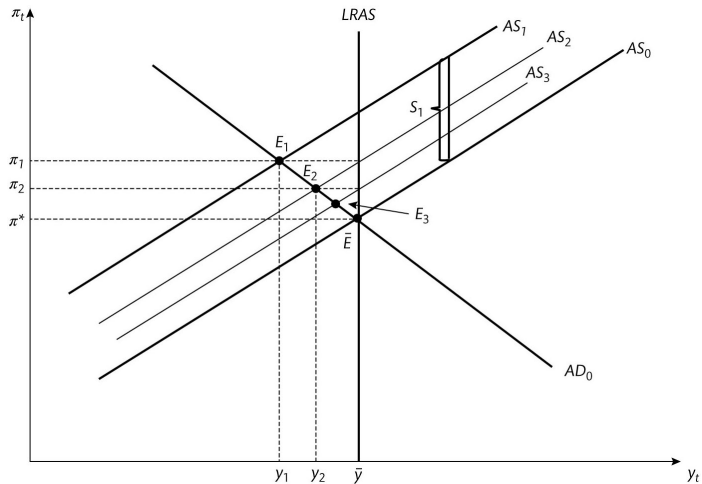
Temporary Supply Shock

Suppose $s_1 > 0$ and $s_t = 0$ for $t \geq 2$, while $z_t = 0$.

$$\hat{\pi}_1 = \beta s_1 > 0, \quad \hat{y}_1 = -a\beta s_1 < 0.$$

- ▶ Inflation rises while output falls.
- ▶ The model predicts negative co-movement.
- ▶ Energy prices, import costs, markups, and productivity disruptions can look like this.
- ▶ This is the useful first-pass map for the 2022 energy-price episode.

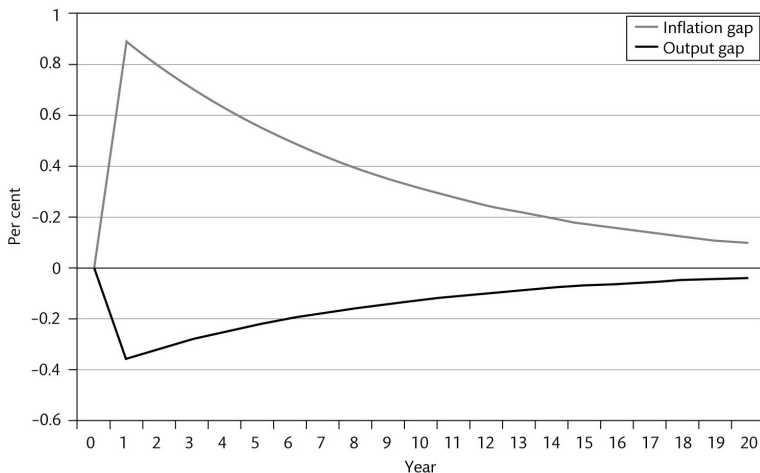
Temporary Supply Shock: Figure



Source: Sorensen & Whitta-Jacobsen, Chapter 19.

A supply impulse shifts AS first and creates the inflation-output trade-off.

Impulse Responses: Supply Shock



Source: Sorensen & Whitta-Jacobsen, Chapter 19.

Inflation declines gradually because static expectations carry the shock forward.

Demand Versus Supply Shocks

	Demand Shock ($z_1 \neq 0$)	Supply Shock ($s_1 \neq 0$)
First curve to shift	AD	AS
Impact on $\hat{y}$	Same sign as z_1	Negative if $s_1 > 0$
Impact on $\hat{\pi}$	Same sign as z_1	Positive if $s_1 > 0$
Co-movement	Output and inflation move together	Output and inflation move apart
Policy dilemma	Avoid overheating or slump	Balance inflation and output losses

Socrative Check

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Question 1: Supply Shock Dynamics

The economy starts in long-run equilibrium. A one-time adverse energy-price shock hits in period 1: $s_1 > 0$, $s_t = 0$ for $t \geq 2$, and $z_t = 0$. With

$$\hat{y}_t = z_t - a\hat{\pi}_t, \quad \hat{\pi}_t = \hat{\pi}_{t-1} + \gamma\hat{y}_t + s_t, \quad \beta = \frac{1}{1 + a\gamma},$$

which statement is correct?

- A. $\hat{\pi}_1 = \beta s_1 > 0$, $\hat{y}_1 = -a\beta s_1 < 0$, and both gaps converge gradually at rate β .
- B. $\hat{\pi}_1 = \gamma\beta s_1 > 0$, $\hat{y}_1 = \beta s_1 > 0$, so output and inflation move together.
- C. $\hat{\pi}_1 = s_1 > 0$, $\hat{y}_1 = 0$, because monetary policy does not react to supply shocks.
- D. $\hat{\pi}_1 = \beta s_1 > 0$, $\hat{y}_1 = -a\beta s_1 < 0$, but inflation is back at target in period 2.
- E. $\hat{\pi}_1 = 0$, $\hat{y}_1 = -as_1 < 0$, because the shock only affects output.

Question 2: Persistence And Policy Strength

Two economies face the same one-time adverse supply shock $s_1 > 0$. They have the same $\gamma > 0$ and b , but economy H has a stronger inflation response in the Taylor-type rule: $h_H > h_L$. Which statement is correct?

$$a = \frac{\alpha_2 h}{1 + \alpha_2 b}, \quad \beta = \frac{1}{1 + a\gamma}$$

- A. Economy H has larger a , smaller β , lower initial inflation, larger initial output loss, and faster convergence.
- B. Economy H has larger a , larger β , higher initial inflation, smaller initial output loss, and slower convergence.
- C. Economy H has smaller a , smaller β , lower initial inflation, smaller initial output loss, and faster convergence.
- D. Economy H has the same a but smaller β , because only γ affects the AD slope.
- E. Economy H has larger a , but this matters only for demand shocks, not for supply shocks.

Question 3: Identifying The Shock

The economy starts in long-run equilibrium. In period 1, output falls below trend and inflation rises above target. A student says: “This must be a pure demand contraction, because lower output means weaker demand.” What is the best assessment?

- A. Correct: any fall in output in the AS–AD model must come from a negative demand shock.
- B. Incorrect: a negative demand shock lowers both output and inflation, while an adverse supply shock raises inflation and lowers output.
- C. Correct only if $a = 0$, because then inflation cannot respond to demand shocks.
- D. Incorrect because supply shocks affect inflation only in the long run, not on impact.
- E. Indeterminate: demand and supply shocks always generate the same signs for output and inflation.

The 2022 Energy Shock Through The Model

Data facts

- ▶ Euro-area HICP inflation averaged 8.4 percent in 2022.
- ▶ Inflation reached rates around 10 percent year-on-year late in 2022.
- ▶ Energy was the largest direct contributor.
- ▶ Denmark had one of the largest household electricity price increases in the EU in the second half of 2022.

Model reading

- ▶ $s_t > 0$: production costs rise.
- ▶ $\hat{\pi}_t > 0$: inflation rises above target.
- ▶ $\hat{y}_t < 0$: activity is squeezed.
- ▶ Policy must choose how much demand compression to add.

Sources: ECB Annual Report 2022; Eurostat electricity and gas price release, 26 April 2023.

Taylor-Type Rule: What Tightening Does

$$i_t^p = \bar{r} + \pi_{t+1}^e + h(\pi_t - \pi^*) + b(y_t - \bar{y})$$

Energy shock ($s_t > 0$) $\Rightarrow \pi_t - \pi^* > 0$ and $y_t - \bar{y} < 0$

$$h(\pi_t - \pi^*) > 0 \Rightarrow i_t^p \uparrow$$

$$b(y_t - \bar{y}) < 0 \Rightarrow i_t^p \downarrow$$

$h \uparrow \Rightarrow$ stronger inflation response $\Rightarrow$ faster disinflation and larger output loss.

Three Policy Stances In The Notebook

Group	AD Parameter	Interpretation
A	$a = 1.50$	Strong inflation response, faster disinflation, larger output loss
B	$a = 0.25$	Softer inflation response, slower disinflation, smaller output loss
C	Choose a	Compromise and defend the choice

$$\hat{\pi}_1 = \beta s_1, \quad \hat{y}_1 = -a\beta s_1, \quad \beta = \frac{1}{1 + a\gamma}.$$

Notebook Group Exercise

Notebook: `L6_as_ad_supply_shocks.ipynb`

- ▶ Work in groups of 2-4.
- ▶ Use $\gamma = 0.5$, $s_1 = 1.0$, and $T = 12$.
- ▶ Group A: use $a = 1.50$ and argue for strong inflation stabilization.
- ▶ Group B: use $a = 0.25$ and argue for limiting the output loss.
- ▶ Group C: use $a = 0.75$ and argue for a compromise.
- ▶ Change the shock from $s_1 = 1.0$ to $s_1 = 2.0$.
- ▶ Prepare one sentence using your impact inflation gap, output gap, and β .

Current-Event Stress Test: Strait Of Hormuz

- ▶ The Strait of Hormuz is one of the world's most important oil transit chokepoints.
- ▶ The IEA reports that about 20 million barrels per day of crude oil and oil products moved through it in 2025.
- ▶ EIA's April 2026 outlook links the Hormuz closure and related production outages to a sharp oil-price increase.
- ▶ EIA forecast Brent to peak around 115 dollars per barrel in 2026 Q2 before easing as outages abate.
- ▶ Recent oil prices: FRED Brent Crude Oil Price, Daily.
- ▶ The model question is the same: temporary supply shock, persistent supply shock, or a risk premium in demand?

Sources: IEA Strait of Hormuz factsheet, February 2026; EIA press release, 7 April 2026.

How To Classify The Iran-Related Oil Shock

If It Is Temporary

- ▶ $s_1 > 0$, then $s_t = 0$.
- ▶ Inflation rises, output falls.
- ▶ Expectations unwind gradually.
- ▶ Policy faces a short-lived trade-off.

If It Is Persistent

- ▶ s_t remains positive for several periods.
- ▶ Inflation pressure lasts longer.
- ▶ Output losses may accumulate.
- ▶ Policy may need to reassess normal output and real-rate conditions.

The hardest part is not drawing AS-AD, it is diagnosing the shock's persistence.

What To Communicate In An Exam Answer

1. State the shock: demand side z_t or supply side s_t .
2. Draw or describe the first curve that shifts.
3. Sign the impact effects on output and inflation.
4. Explain propagation through static expectations.
5. Discuss the policy choice using a , h , and the Taylor-type rule.
6. Separate temporary from persistent shocks.

A good answer connects equations, a diagram, and the economic episode.

Takeaways

- ▶ The AS-AD model combines demand, supply, expectations, and policy into one dynamic system.
- ▶ Static expectations make shocks persistent.
- ▶ $\beta = 1/(1 + a\gamma)$ summarizes the speed of convergence.
- ▶ Demand shocks move output and inflation in the same direction on impact.
- ▶ Supply shocks move output and inflation in opposite directions on impact.
- ▶ Energy-price shocks are useful cases because they make the policy trade-off visible.
- ▶ The 2022 euro-area shock and the 2026 Hormuz stress test can be read through the same model.

Preview Of Next Lecture

- ▶ We move from a single shock episode to repeated business-cycle fluctuations.
- ▶ The same propagation logic helps explain why economies do not return instantly to normal.
- ▶ We will connect model dynamics to stabilization policy and uncertainty.
- ▶ The exam skill is to classify a shock, trace the mechanism, and defend the policy trade-off.

Extra Derivations And Permanent Shocks

Appendix A: AD Derivation

Start from the Fisher equation and the policy rule:

$$r_t - \bar{r} = (i_t - \pi_{t+1}^e) - \bar{r} = h(\pi_t - \pi^*) + b(y_t - \bar{y}) + \hat{\rho}_t.$$

Insert this into IS:

$$y_t - \bar{y} = \alpha_1(g_t - \bar{g}) - \alpha_2\{h(\pi_t - \pi^*) + b(y_t - \bar{y}) + \hat{\rho}_t\} - \alpha_3(\tau_t - \bar{\tau}) + v_t.$$

Collect output terms:

$$(1 + \alpha_2 b)(y_t - \bar{y}) = \alpha_1(g_t - \bar{g}) - \alpha_3(\tau_t - \bar{\tau}) + v_t - \alpha_2 \hat{\rho}_t - \alpha_2 h(\pi_t - \pi^*).$$

Divide by $1 + \alpha_2 b$:

$$y_t - \bar{y} = z_t - a(\pi_t - \pi^*)$$

Appendix B: Output-Gap Recursion

Start from

$$\hat{y}_t = z_t - a\hat{\pi}_t, \quad \hat{\pi}_t = \beta\hat{\pi}_{t-1} + \gamma\beta z_t + \beta s_t.$$

Use $\hat{\pi}_{t-1} = (z_{t-1} - \hat{y}_{t-1})/a$:

$$\hat{y}_t = z_t - a \left(\beta \frac{z_{t-1} - \hat{y}_{t-1}}{a} + \gamma\beta z_t + \beta s_t \right).$$

Since $1 - a\gamma\beta = \beta$,

$$\boxed{\hat{y}_t = \beta\hat{y}_{t-1} + \beta(z_t - z_{t-1}) - a\beta s_t}$$

Appendix C: Half-Life Of A Shock

Without new shocks,

$$x_t = \beta x_{t-1}.$$

The half-life $T_{1/2}$ solves

$$\beta^{T_{1/2}} = \frac{1}{2}.$$

Therefore

$$T_{1/2} = \frac{\ln(1/2)}{\ln(\beta)}$$

- ▶ If β is close to 1, shocks decay slowly.
- ▶ If β is small, shocks decay quickly.
- ▶ The notebook lets you compare half-lives across policy responses.

Appendix D: Permanent Supply Shock

If a supply shock is permanent, $s_t = s > 0$, the long-run output reference changes:

$$0 = \gamma(y - \bar{y}) + s \quad \Rightarrow \quad \bar{y}_0 = \bar{y} - \frac{s}{\gamma}.$$

The real rate consistent with goods-market clearing also changes:

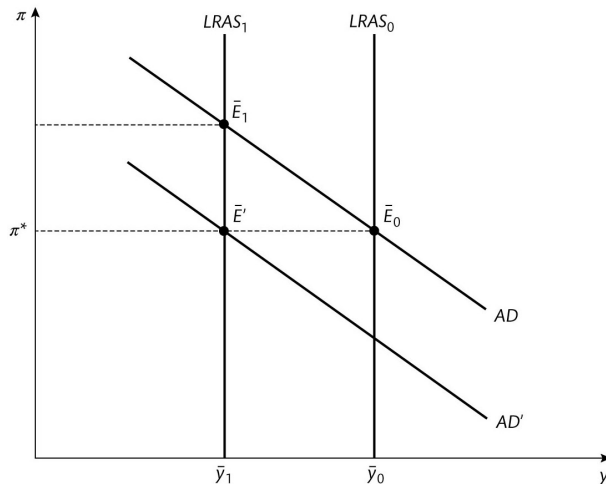
$$\bar{r}_0 = \bar{r} + \frac{s}{\gamma\alpha_2}.$$

A re-centred Taylor-type rule is

$$i_t^p = \bar{r}_0 + \pi_{t+1}^e + h(\pi_t - \pi^*) + b(y_t - \bar{y}_0).$$

Permanent supply shocks require a reassessment of normal output, not only a rate response.

Appendix E: Permanent Supply Shock Figure



Source: Sorensen & Whitta-Jacobsen, Figure 19.9.

If normal output is lower, policy must be centred on the new long-run equilibrium.